SUN'S DIVISIBILITY CONJECTURE FOR $\sum_{k<n}(20k+17)\,W_k(1,-4)$ IS TRUE

QEEDDD VERIFIABLE-MATHEMATICAL-REASONING-CORPUS PROJECT

Abstract. Let $a_n=\sum_k(-4)^k\binom{n}{k}^2\binom{n-k}{k}^2=W_n(1,-4)$ and $S(n)=\sum_{k=0}^{n-1}(20k+17)a_k$. In 2010 Z.-W. Sun conjectured that $n\mid S(n)$ for every $n\ge 1$ (OEIS A179524). We prove this conjecture. Earlier work established the 2-part, the 5-part and all prime powers p^e with $p\equiv 11,13,17,19 \pmod{20}$. The split primes $p\equiv 1,3,7,9 \pmod{20}$ remained open. At these primes the complex-multiplication elliptic curve behind the problem has ordinary reduction, and none of the earlier vanishing-moment mechanisms applies. We treat them with the unit-root crystals of Beukers and Vlasenko. First, an exact identity reduces $S(N) \bmod N$ to the constant term of $\Lambda^N\rho\Phi(uv)$. This reduces the p-part of the conjecture to a single p-adic number: the value at $b=-4$ of the Gauss–Manin connection η_b on a rank-one unit-root crystal. Our main new result identifies this connection: $\eta_b(b)=-\frac12-\eta_{\mathcal{R}}(1/(256b))$, where $\eta_{\mathcal{R}}$ is the unit-root connection of Ramanujan's family ${}_3F_2(\frac14,\frac12,\frac34;1,1;256T)$. The proof expands a second model of the crystal at a corner of its Newton polytope; the resulting Hasse–Witt coefficients converge p-adically to Ramanujan's period $c^{-1}\,{}_3F_2(\frac14,\frac12,\frac34;1,1;1/c^2)$. Sun's theorem on the partial sums of Ramanujan's series $\sum(20k+3)\binom{2k}{k}^2\binom{4k}{2k}(-2^{10})^{-k}=8/\pi$ then gives $20\eta_b(-4)+7=0$, which yields $p^{v_p(N)}\mid S(N)$ for every N. All steps are unconditional; an exact-arithmetic verification script is included.

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1. Introduction

For $n \geq 0$ put

$$M(n,k) = \binom{n}{k}\binom{n-k}{k} = \binom{n}{2k}\binom{2k}{k}, \qquad a_n = \sum_{0 \leq 2k \leq n} (-4)^k M(n,k)^2,$$

and

$$S(n) = \sum_{k=0}^{n-1} (20k+17) a_k.$$

In Sun's notation $a_n = W_n(1,-4)$, where $W_n(b,c) = \sum_k \binom{n}{2k}^2 \binom{2k}{k}^2 b^{n-2k} c^k$. The first values are $a_0, a_1, a_2, a_3 = 1, 1, -15, -143$ and $S(1), \ldots, S(4) = 17, 54, -801, -11812$.

Conjecture (Z.-W. Sun, 2010) ([16, Conjecture 5.8(ii)], [12])**.** *For every positive integer n we have $n \mid S(n)$.*

The conjecture was stated in the eleventh version of [16] (dated 1 July 2010 in OEIS A179524 [12]). It is recorded as an open statement in the Lean database `formal-conjectures` [9] and in the *OEIS Open* benchmark [13].

Main Theorem. *For every integer $n \geq 1$, $n \mid S(n)$. Moreover $v_2(S(n)) = v_2(n)$.*

1.1. What was known.

The following facts are reproved below for completeness. (i) $\mathrm{rad}(n) \mid S(n)$ for every n, by a Lucas congruence and a reflection that turns $S(p) \bmod p$ into a truncation of Ramanujan's series [7, 10]. (ii) $S(n)/n$ is an odd 2-adic integer [10]. (iii) $p^e \mid S(p^e m)$ for all $e, m \geq 1$ when $p = 5$ or $p \equiv 11, 13, 17, 19 \pmod{20}$ [10]. This uses an exact "digit operator" and the vanishing of three moments, which comes from the supersingular reduction of the CM curve $j = j(\sqrt{-5})$. At the *split primes*, $p \equiv 1, 3, 7, 9 \pmod{20}$, the CM curve is ordinary and the moments are units. Before this paper only finitely many prime-power levels were known there, by computation [7, 10].

1.2. The new result.

Theorem 1.1. *Let p be a prime with $p \equiv 1, 3, 7, 9 \pmod{20}$. Then $p^{v_p(N)} \mid S(N)$ for every $N \geq 1$.*

The Main Theorem follows from Theorem 1.1 together with Theorems 3.1, 5.5 and 5.6 (Section 10).

1.3. Strategy for the split primes.

(1) *An exact reduction* (Section 2). A telescoping identity gives $S(N) = 10N \cdot (\text{integer}) + H_2(N)$, and $H_2(N) = \mathrm{CT}(\Lambda^N \rho \Phi(uv))$ with $\Lambda = (1+u)(1+v)/(uv)$, $\rho = uv/(1+u+v)$ and $\Phi(x) = \sum_l (20l+7)(-4)^l \binom{2l}{l}^2 x^{2l}$. So it suffices to show that the power series $\rho\Phi(uv)$ is a p-adic *formal divergence* $\theta_u J_1 + \theta_v J_2$.

(2) *A rank-one unit-root crystal* (Section 7). The series $\rho\Phi(uv)$ is a constant term of the rational function $\Omega = (20\theta_b + 7)\big(uv/f_b\big)$ at $b = -4$, where $f_b = (1+u+v)(1-uv(z+b/z)(w+1/w))$. At an ordinary prime, the Beukers–Vlasenko theory of Dwork crystals [2] gives $\Omega \equiv (20\eta_b + 7)\, uv/f_b$ modulo formal derivatives, with a unit-root connection η_b. So everything reduces to the single identity $20\eta_b(-4) + 7 = 0$.

(3) *Identification with Ramanujan's family* (Section 9). We prove $\eta_b(b) = -\frac{1}{2} - \eta_{\mathcal{R}}(1/(256b))$, where $\eta_{\mathcal{R}} = \theta_T \log \mathcal{F}(T)$ and $\mathcal{F}(T) = \sum_k \frac{(4k)!}{k!^4} T^k$. To prove it we use a second model $f_c = (1 - cxg(z,w))(1 - xL(y))$ of the same crystal ($b = c^2$) and expand it at a *corner* vertex of its Newton polytope. There the Hasse–Witt coefficients are explicit binomial sums that converge p-adically, as the level grows, to Ramanujan's period $c^{-1}\mathcal{F}(1/(256c^2))$.

(4) *Sun's theorem* (Section 8). Sun proved via a WZ pair that $4(2n+1)\binom{2n}{n}$ divides the partial sums of Ramanujan's $(20k+3)$-series [15, Theorem 1.2]. Together with Dwork's congruences [11, 3] this gives $\eta_{\mathcal{R}}(-1/1024) = -3/20$. Hence $20\eta_b(-4) + 7 = 20(-\frac{1}{2} + \frac{3}{20}) + 7 = 0$.

Remark 1.2 (An upside-down Ramanujan formula). The "period" of the crystal in step (2) is $\int_0^{1/4} {}_2F_1(\frac{1}{2},\frac{1}{2};1;16bx^2)\,(1-4x)^{-1/2}\,dx = \frac{1}{2}\,{}_3F_2(\frac{1}{2},\frac{1}{2},\frac{1}{2};\frac{3}{4},\frac{5}{4};b)$, a series that diverges at $b = -4$. The point of maximal unipotent monodromy of its differential equation is $b = \infty$, where the holomorphic solution is $b^{-1/2}{}_3F_2(\frac{1}{4},\frac{1}{2},\frac{3}{4};1,1;1/b)$. Formally, Sun's sum is the p-adic shadow of the "upside-down" series $\sum_l (20l+7)(-4)^l \frac{(1/2)_l^3}{(1)_l(3/4)_l(5/4)_l}$. Up to sign, this is Ramanujan's series $\sum_k (20k+3)\frac{(1/2)_k(1/4)_k(3/4)_k}{k!^3}(-\frac{1}{4})^k = \frac{8}{\pi}$ continued to the half-integers $k = -l - \frac{1}{2}$. Theorem 9.1 makes this precise p-adically, at every ordinary prime.

Conventions. $\theta_y = y\,\partial/\partial y$ for any variable y. CT denotes the constant term. v_p is the p-adic valuation, with $v_p(0) = \infty$. $(a)_k$ is the rising factorial. For an odd prime p we write $h = (p-1)/2$ and $d = \lfloor p/4 \rfloor$. We put $\gamma_l = (-4)^l\binom{2l}{l}^2$, $A_k = \frac{(4k)!}{k!^4} = \binom{2k}{k}^2\binom{4k}{2k}$, and

$$G_b(x) = \sum_{l\ge 0} b^l \binom{2l}{l}^2 x^{2l}, \qquad E(x) = \sum_{i\ge 0}\binom{2i}{i}x^i = (1-4x)^{-1/2}, \qquad \Phi(x) = \sum_{l\ge 0}(20l+7)\gamma_l x^{2l}.$$

Thus $G_{-4}(x) = \sum_l \gamma_l x^{2l}$ and $\Phi = \big((20\theta_b + 7)G_b\big)|_{b=-4}$. For $N \ge 1$ we put

$$d_N(b) = [x^{N-1}]\,G_b(x)E(x) = \sum_{0\le 2l\le N-1} b^l \binom{2l}{l}^2\binom{2i}{i}, \qquad i = N-1-2l. \tag{1}$$

2. Elementary reductions

Lemma 2.1. *For integers $m, N \ge 0$, $\displaystyle\sum_{n=0}^{N-1}(2n-m+1)\binom{n}{m}^2 = N\binom{N}{m+1}\binom{N-1}{m}$ (the right side is read as 0 for $N = 0$).*

Proof. Put $g(n) = (n-m)^2\binom{n}{m}^2/(m+1)$. For $n \ge 1$ this equals $n\binom{n}{m+1}\binom{n-1}{m}$, and $g(0) = 0$. Since $\binom{n+1}{m} = \frac{n+1}{n+1-m}\binom{n}{m}$ for $n \ge m$,

$$g(n+1) - g(n) = \binom{n}{m}^2 \frac{(n+1)^2 - (n-m)^2}{m+1} = (2n-m+1)\binom{n}{m}^2.$$

This holds for all $n \ge 0$; for $n < m$ both sides vanish. Now sum over $0 \le n < N$. □

Proposition 2.2. *Let*

$$H_3(N) = \sum_{n<N}\sum_k (2n-2k+1)\gamma_k\binom{n}{2k}^2, \qquad H_2(N) = \sum_{n<N}\sum_k (20k+7)\gamma_k\binom{n}{2k}^2.$$

Then $S(N) = 10H_3(N) + H_2(N)$ and $H_3(N) = N\sum_k \gamma_k\binom{N}{2k+1}\binom{N-1}{2k}$. In particular $S(N) \equiv H_2(N) \pmod N$.

Proof. We have $a_n = \sum_k \gamma_k\binom{n}{2k}^2$ and $10(2n-2k+1) + (20k+7) = 20n+17$. Apply Lemma 2.1 with $m = 2k$. □

Proposition 2.3. *Let $\Lambda = (1+u)(1+v)/(uv)$ and $\rho = uv/(1+u+v) \in \mathbb{Z}[[u,v]]$. For every $R(x) = \sum_l r_l x^{2l} \in \mathbb{Q}[[x]]$ and $N \geq 1$,*

$$\operatorname{CT}\big(\Lambda^N \rho\, R(uv)\big) = \sum_{n<N} \sum_{l} r_l \binom{n}{2l}^2,$$

the constant term being taken in $(uv)^{-N}\mathbb{Q}[[u,v]]$. In particular $H_2(N) = \operatorname{CT}(\Lambda^N \rho \Phi(uv))$.

Proof. We have $\Lambda - 1 = (1+u+v)/(uv) = \rho^{-1}$, so $\Lambda^N \rho = \sum_{n<N} \Lambda^n + \rho$. Moreover $\operatorname{CT}(\rho R(uv)) = 0$, because $\rho \in uv\mathbb{Z}[[u,v]]$. Finally $\operatorname{CT}(\Lambda^n (uv)^{2l}) = [u^{n-2l}](1+u)^n\,[v^{n-2l}](1+v)^n = \binom{n}{2l}^2$. □

Lemma 2.4 (Divergence criterion)**.** *Let p be a prime and $\Psi = \sum_{(r,s)\neq(0,0)} \psi_{rs} u^r v^s \in \mathbb{Z}_p[[u,v]]$ with $\psi_{rs} \in p^{\min(v_p(r), v_p(s))}\mathbb{Z}_p$ for all (r,s). Then $\operatorname{CT}(\Lambda^N \Psi) \in N\mathbb{Z}_p$ for every $N \geq 1$.*

Proof. For $(r,s) \neq (0,0)$ choose $x_{rs}, y_{rs} \in \mathbb{Z}_p$ with $x_{rs} r + y_{rs} s = p^{\min(v_p r, v_p s)}$; this is Bézout in the discrete valuation ring $\mathbb{Z}_p$. Then $\Psi = \theta_u J_1 + \theta_v J_2$, where $J_1, J_2 \in \mathbb{Z}_p[[u,v]]$ have coefficients $\psi_{rs} x_{rs} p^{-\min}$ and $\psi_{rs} y_{rs} p^{-\min}$. Since $\operatorname{CT}(\theta_u(\cdot)) = 0$,

$$\operatorname{CT}(\Lambda^N \theta_u J_1) = -\operatorname{CT}(\theta_u(\Lambda^N) J_1) = -N\, \operatorname{CT}\big(\Lambda^{N-1} \theta_u(\Lambda) J_1\big) \in N\mathbb{Z}_p,$$

because $\theta_u \Lambda = -(1+v)/(uv)$ has integer coefficients. The v-term is the same. □

Corollary 2.5. *Let p be a prime. Suppose $[u^r v^s]\,\rho\Phi(uv) \in p^{\min(v_p(r), v_p(s))}\mathbb{Z}_p$ for all $r, s \geq 0$. Then $p^{v_p(N)} \mid S(N)$ for all $N \geq 1$.*

Proof. Combine Propositions 2.2 and 2.3 with Lemma 2.4. □

We record two more identities used in Sections 3 and 4:

$$\binom{j}{m}^2 = \sum_{i=0}^{m} \binom{m}{i}\binom{m+i}{i}\binom{j}{m+i}, \tag{2}$$

$$\sum_{j<n} (20j+17)\binom{j}{t} = (20t+17)\binom{n}{t+1} + 20(t+1)\binom{n}{t+2}. \tag{3}$$

The first counts ordered pairs of m-subsets of a j-set by the size of their union. The second follows from the hockey-stick identity and $j\binom{j}{t} = t\binom{j}{t} + (t+1)\binom{j}{t+1}$.

3. The prime 2

The following argument is due to [10, Theorem 3.1].

Theorem 3.1. *For every $n \geq 1$, $S(n)/n$ is an odd 2-adic integer. In particular $v_2(S(n)) = v_2(n)$.*

Proof. Using (2) with $m = 2k$ and then (3), we get, with $t = 2k+i$,

$$\frac{S(n)}{n} = \sum_k \gamma_k \sum_{i=0}^{2k} \binom{2k}{i}\binom{2k+i}{i}\Big(\frac{20t+17}{t+1}\binom{n-1}{t} + \frac{20(t+1)}{t+2}\binom{n-1}{t+1}\Big),$$

since $\binom{n}{t+1}/n = \binom{n-1}{t}/(t+1)$ and $\binom{n}{t+2}/n = \binom{n-1}{t+1}/(t+2)$. For $k \geq 1$, $v_2(\gamma_k) \geq 2k+2$ because $\binom{2k}{k}$ is even. Also $t+2 \leq 4k+2 < 2^{2k+2}$, so every term with $k \geq 1$ lies in $2\mathbb{Z}_2$. The term $k = 0$ equals $n^{-1}\sum_{j<n}(20j+17) = 10n+7$, which is odd. □

4. The Hasse invariant and the CM dichotomy

For an odd prime p and an indeterminate b put $U_p(b)=\sum_{r<p}\sum_k b^k\binom{r}{2k}^2\binom{2k}{k}^2$. Thus $U_p(-4)=\sum_{r<p}a_r$. Let P_d be the Legendre polynomial.

Proposition 4.1. *Let p be an odd prime. In $\mathbb{F}_p[b,b^{-1}]$,*

$$d_p(b)\equiv U_p(b)\equiv b^h\sum_{j=0}^{d}A_j(256b)^{-j}\equiv b^hP_d(t)^2,\qquad t^2=1-\frac{1}{b}.$$

Here $P_d(t)^2$ is a polynomial in t^2, so the right side lies in $\mathbb{F}_p[b,b^{-1}]$.

Proof. By (2) and $\sum_{r<p}\binom{r}{m+i}=\binom{p}{m+i+1}$, only $i=p-1-m$ survives modulo p. Hence $\sum_{r<p}\binom{r}{2k}^2\equiv\binom{2k}{p-1-2k}$, because $\binom{p-1}{i}\equiv(-1)^i=1$. Therefore $U_p(b)\equiv\sum_k b^k\binom{2k}{k}^2\binom{2k}{p-1-2k}$.

Put $k=h-j$ in U_p and $l=h-j$ in (1). The coefficients of b^{h-j} are $\binom{2h-2j}{h-j}^2\binom{p-1-2j}{2j}$ and $\binom{2h-2j}{h-j}^2\binom{4j}{2j}$ respectively. For $j\le d$ we have $\binom{p-1-2j}{2j}\equiv\binom{-1-2j}{2j}=\binom{4j}{2j}$, since $(2j)!$ is a p-unit. For $d<j\le h$ both coefficients vanish modulo p: $4j>p-1$ forces a carry in $2j+2j$, and $\binom{p-1-2j}{2j}=0$. So $d_p\equiv U_p$.

Next, for $j\le d$, $\binom{2h-2j}{h-j}\equiv\binom{-1-2j}{h-j}=(-1)^{h-j}\binom{h+j}{2j}$ and, since $h\equiv-\frac12$, $\binom{h+j}{2j}\equiv\binom{j-1/2}{2j}=(-1)^j\binom{2j}{j}16^{-j}$. Hence $\binom{2h-2j}{h-j}^2\equiv\binom{2j}{j}^2256^{-j}$, which gives the third expression.

Finally, $A_j256^{-j}=(\frac14)_j(\frac12)_j(\frac34)_j/j!^3$. Also $\{-d,d+1\}\equiv\{\frac14,\frac34\}$, so $(\frac14)_j(\frac34)_j\equiv(-d)_j(d+1)_j$ for $j\le d$. The classical identity $P_d(t)^2={}_3F_2(-d,d+1,\frac12;1,1;1-t^2)=\sum_j\binom{d}{j}\binom{d+j}{j}\binom{2j}{j}\left(\frac{t^2-1}{4}\right)^j$ [1] then gives the last congruence with $1-t^2=1/b$. □

Proposition 4.2. *Let p be an odd prime. Then $d_p(-4)\not\equiv0\pmod p$ if and only if $p\equiv1,3,7,9\pmod{20}$.*

Proof. For $p=5$, $d_5(-4)=550\equiv0$. Let $p\ne5$. By Proposition 4.1 at $b=-4$, $d_p(-4)\equiv(-4)^hP_d(t_0)^2$ with $t_0^2=\frac54$, $t_0\in\overline{\mathbb{F}}_p$. Let $t_0=\sqrt5/2$ and consider $E:\ y^2=x\bigl(x^2+2x+\frac{1-t_0}{2}\bigr)$ over $\mathbb{Q}(\sqrt5)$. Its discriminant $\Delta=8(t_0-1)^2(t_0+1)$ is a unit at every prime above an odd rational prime, because $(t_0-1)(t_0+1)=\frac14$. Its j-invariant is $64(3t_0+5)^3/((t_0-1)^2(t_0+1))=632000+282880\sqrt5=j(\sqrt{-5})$ [6], so E has complex multiplication by $\mathbb{Z}[\sqrt{-5}]$, of discriminant -20.

The Hasse invariant of the reduction is the coefficient of x^{p-1} in $(x^3+2x^2+b'x)^h$, with $b'=\frac{1-t_0}{2}$ [14, Theorem V.4.1(a)]. Choosing k factors $b'x$, $h-2k$ factors $2x^2$ and k factors x^3, this coefficient is $2^h\sum_k\frac{(-h/2)_k((1-h)/2)_k}{k!^2}b'^k$. Since $h\equiv-\frac12$, we have $\{-\frac h2,\frac{1-h}{2}\}\equiv\{\frac14,\frac34\}\equiv\{-d,d+1\}$, and the sum terminates at $k=d$. So the coefficient is $\equiv2^h{}_2F_1(-d,d+1;1;b')=2^hP_d(1-2b')=2^hP_d(t_0)$ (cf. [10, §5]; checked in the Appendix). By Deuring's reduction theorem [5, 8], the reduction at a prime above $p\nmid20$ is ordinary if and only if p splits in $\mathbb{Q}(\sqrt{-5})$, i.e. $\left(\frac{-20}{p}\right)=1$, i.e. $p\equiv1,3,7,9\pmod{20}$. □

Corollary 4.3. *Let $p\equiv1,3,7,9\pmod{20}$ and $t_0=-1/1024$. Then $d_p(-4)$, $U_p(-4)$ and $\mathcal{F}_{<p}(t_0)=\sum_{k<p}A_kt_0^k$ are p-adic units.*

Proof. $A_k\equiv0\pmod p$ for $d<k<p$, so $\mathcal{F}_{<p}(t_0)\equiv\sum_{k\le d}A_k(-1024)^{-k}$. This is $(-4)^{-h}d_p(-4)$ modulo p by Proposition 4.1. □

Lemma 4.4 (Lucas factorisation)**.** *For every odd prime p and $s\ge1$, $d_{p^s}(b)\equiv d_p(b)^{(p^s-1)/(p-1)}\pmod p$ in $\mathbb{Z}[b]$.*

Proof. Write $l = l_0 + pl'$ with $0 \le l_0 < p$. Lucas' theorem gives $\binom{2l}{l} \equiv \binom{2l_0}{l_0}\binom{2l'}{l'} \pmod p$; when $2l_0 \ge p$ both sides vanish. In $\mathbb{F}_p[b]$ we also have $b^l = b^{l_0}(b^{l'})^p$. Hence $G_b(x) \equiv G_{b,<p}(x)\, G_b(x)^p$ in $\mathbb{F}_p[b][[x]]$, where $G_{b,<p} = \sum_{2l_0 \le p-1} b^{l_0}\binom{2l_0}{l_0}^2 x^{2l_0}$. Likewise $E(x) \equiv E_{<p}(x)E(x)^p$ with $E_{<p} = \sum_{2i \le p-1}\binom{2i}{i}x^i$. So $\Psi := G_b E \equiv \Psi_1 \Psi^p$ with $\Psi_1 = G_{b,<p}E_{<p}$ of degree $< 2p-1$. Since Ψ^p is a series in x^p, the coefficient of x^{p^s-1} in $\Psi_1\Psi^p$ comes only from x^{p-1} in Ψ_1. As $[x^{p-1}]\Psi_1 = [x^{p-1}]\Psi = d_p(b)$, we get $d_{p^s} \equiv d_p\, d_{p^{s-1}}^{\,p}$, and the claim follows by induction on s. □

5. The primes 5 and $p \equiv 11, 13, 17, 19 \pmod{20}$

This section reproduces, with complete proofs, the exact digit operator of [10, §§5–6]. Fix a prime $p \ge 5$ and put $w(n,k) = (-4)^k M(n,k)^2$ for $0 \le 2k \le n$, and $w(n,k) = 0$ otherwise. Let $\mathcal{A}_p = \mathbb{Z}_p\langle X, Y\rangle$ be the ring of restricted power series. Its coefficients tend to 0, so its elements can be evaluated on $\mathbb{Z}_p^2$. Define

$$I_N(F) = \sum_{n<N}\sum_{k} w(n,k)F(n,k), \qquad U(N) = I_N(1),\ T(N) = I_N(X),\ V(N) = I_N(Y),$$

so $S(N) = I_N(20X+17)$, $U(N) = \sum_{n<N} a_n$ and $T(N) = \sum_{n<N} n a_n$. Put $W = (p-1)!$, $E_p(N) = \prod_{1\le i\le N,\, p\nmid i} i$, $H_r = \sum_{i\le r} 1/i$ and $Q = ((-4)^{p-1}-1)/p \in \mathbb{Z}_p$.

Lemma 5.1. *There are $\mathcal{H}_r \in 1 + p\mathbb{Z}_p\langle X\rangle$ $(0 \le r < p)$ with $E_p(pq+r) = W^q r!\, \mathcal{H}_r(q)$ for all integers $q \ge 0$, and $\mathcal{H}_r(X) \equiv 1 + pH_r X \pmod{p^2}$.*

Proof. Let $B(X) = \prod_{i=1}^{p-1}(1 + pX/i)$. Then $\log B(X) = \sum_{j\ge1} \ell_j X^j$ with $\ell_j = (-1)^{j-1}p^j\eta_j/j$, where $\eta_j = \sum_{i<p} i^{-j}$. Let $S_j(X) = \sum_{a=0}^{j}\left\{{j \atop a}\right\}(X)_{\underline{a+1}}/(a+1)$ be the polynomial with $S_j(q) = \sum_{h<q} h^j$. Its coefficients have valuation $\ge -\lfloor \log_p(j+1)\rfloor$. Put $\mathcal{A}(X) = \sum_j \ell_j S_j(X)$. By Wolstenholme, $v_p(\eta_1) \ge 2$ and $v_p(\eta_2) \ge 1$ for $p \ge 5$; for $j \ge 3$ we have $j - v_p(j) - \lfloor \log_p(j+1)\rfloor \ge 3$, with a bound tending to ∞. Hence $\mathcal{A} \in p^3\mathbb{Z}_p\langle X\rangle$ and $\mathcal{E} = \exp\mathcal{A} \in 1 + p^3\mathbb{Z}_p\langle X\rangle$. At integers $q \ge 0$ we get $\mathcal{E}(q) = \prod_{h<q} B(h)$. Set $\mathcal{H}_r(X) = \mathcal{E}(X)\prod_{i\le r}(1 + pX/i)$. Then $E_p(pq+r) = \prod_{h<q}\prod_{i<p}(ph+i)\cdot \prod_{i\le r}(pq+i) = W^q r!\, \mathcal{H}_r(q)$. □

Lemma 5.2. *For $0 \le r, s < p$ let $\varepsilon = \lceil (2s-r)/p\rceil \in \{0,1,2\}$, $t = r - 2s + \varepsilon p \in [0,p)$ and $z = X - 2Y$. Put*

$$K_{rs} = p^{2\varepsilon}(z)_{\underline{\varepsilon}}^2 \frac{(-4)^s W^{2\varepsilon} r!^2}{s!^4 t!^2}\frac{\mathcal{C}(Y)\mathcal{H}_r(X)^2}{\mathcal{H}_s(Y)^4\mathcal{H}_t(X-2Y-\varepsilon)^2} \in \mathcal{A}_p, \qquad \mathcal{C}(Y) = \exp\bigl(Y\log((-4)^{p-1})\bigr).$$

Then $w(pq+r, pj+s) = w(q,j)K_{rs}(q,j)$ for all $q, j \ge 0$, and $K_{rs}(0,0) = w(r,s)$. If $\varepsilon \ge 1$ then $K_{rs} \in p^2\mathcal{A}_p$. If $\varepsilon = 0$ then $K_{rs} \equiv w(r,s)\bigl[1 + p\{2X(H_r - H_t) + Y(Q + 4H_t - 4H_s)\}\bigr] \pmod{p^2}$.

Proof. Use $N! = p^{\lfloor N/p\rfloor}\lfloor N/p\rfloor!\, E_p(N)$ and Lemma 5.1 for $n = pq+r$, $k = pj+s$ and $n - 2k = p(q-2j-\varepsilon) + t$. The powers of W cancel down to W^ε, and $q_0!/(q_0-\varepsilon)! = (q_0)_{\underline{\varepsilon}}$ with $q_0 = q - 2j$. Also $(-4)^{pj+s} = (-4)^{j+s}\mathcal{C}(j)$. If $0 \le q_0 < \varepsilon$, both sides vanish, and if $2j > q$ both sides are 0. The denominators are units of $\mathcal{A}_p$, since affine integral substitutions preserve $1 + p\mathcal{A}_p$. The remaining claims follow from $\mathcal{C} \equiv 1 + pQY$ and $\mathcal{H}_r \equiv 1 + pH_rX \pmod{p^2}$. □

Proposition 5.3. *For $F \in \mathcal{A}_p$ put $(\mathcal{T}F)(X,Y) = \sum_{r,s<p} K_{rs}(X,Y)F(pX+r, pY+s)$. Then $I_{pN}(F) = I_N(\mathcal{T}F)$ for $N \ge 1$. Moreover there are $\alpha_F, \beta_F \in \mathbb{Z}_p$ and $R_F \in \mathcal{A}_p$ with $\mathcal{T}F = I_p(F) + p\alpha_F X + p\beta_F Y + p^2 R_F$, so that*

$$I_{pN}(F) = I_p(F)U(N) + p\alpha_F T(N) + p\beta_F V(N) + p^2 I_N(R_F). \tag{4}$$

Proof. Split $n = pq+r$ and $k = pj+s$ and apply Lemma 5.2. For the second statement use $F(pX+r, pY+s) \equiv F(r,s) + pXF_X(r,s) + pYF_Y(r,s) \pmod{p^2}$, which holds for restricted series by Taylor's formula with divided derivatives. The constant term is $\sum_{r,s} w(r,s)F(r,s) = I_p(F)$. □

Theorem 5.4. *Suppose $U(p), T(p), V(p) \in p\mathbb{Z}_p$. Then for all $e, m \geq 1$ and $F \in \mathcal{A}_p$ we have $I_{p^e m}(F) \in p^{e-1}\mathbb{Z}_p$, and $I_{p^e m}(F) \in p^e\mathbb{Z}_p$ when $I_p(F) \in p\mathbb{Z}_p$.*

Proof. Induct on e. The case $e = 1$ follows from (4) with $N = m$. For $e \geq 2$ put $N = p^{e-1}m$. By the strong assertion at $e-1$, applied to $F = 1, X, Y$ (whose I_p are $U(p), T(p), V(p) \in p\mathbb{Z}_p$), we get $U(N), T(N), V(N) \in p^{e-1}\mathbb{Z}_p$. By the weak assertion, $I_N(R_F) \in p^{e-2}\mathbb{Z}_p$. Now read off (4). □

Theorem 5.5. *If $p \equiv 11, 13, 17, 19 \pmod{20}$, then $p^e \mid S(p^e m)$ for all $e, m \geq 1$.*

Proof. By Propositions 4.1 and 4.2, $U(p) = U_p(-4) \equiv (-4)^h P_d(t_0)^2 = 0$. Next, $V(p) = (\theta_b U_p)(-4)$. Differentiating the polynomial congruence $U_p(b) \equiv b^h P_d(t)^2$ and using $P_d(t_0) = 0$ and $t_0 \neq 0$ gives $V(p) \equiv 0$. Finally, $F_k(X) = \binom{X}{2k}^2$ satisfies $F_k(2k-1-X) = F_k(X)$ over $\mathbb{F}_p$. Summing over $X \in \mathbb{F}_p$ gives $\sum_{r<p} rF_k(r) \equiv (k - \frac{1}{2})\sum_r F_k(r)$, hence $T(p) \equiv V(p) - \frac{1}{2}U(p) \equiv 0$. Apply Theorem 5.4 with $F = 20X + 17$, using $I_p(F) = S(p) = 20T(p) + 17U(p)$. □

Theorem 5.6. *For all $e, m \geq 1$, $5^e \mid S(5^e m)$.*

Proof. Take $p = 5$. From Lemma 5.2, $\mathcal{T}(1) = U(5) + 5\alpha X + 5\beta Y + 25R$ with $\alpha = \sum 2w(r,s)(H_r - H_t)$ and $\beta = \sum w(r,s)(Q + 4H_t - 4H_s)$, the sums over pairs with $\varepsilon = 0$. Exact evaluation gives $U(5) = -155$, $\alpha = 1440$ and $\beta = -37270/3$, all divisible by 5 (see the Appendix). Hence $\mathcal{T}(1) = C + 25R$ with $C \in 5\mathbb{Z}_5$. Now we prove, by induction on e: (a) $I_{5^e m}(F) \in 5^{e-1}\mathbb{Z}_5$ for all F; (b) $U(5^e m) \in 5^e\mathbb{Z}_5$. For $e = 1$, (a) is integrality and (b) is $U(5m) = CU(m) + 25I_m(R)$. For the step, use (4) for (a). For (b), use $U(5N) = CU(N) + 25I_N(R)$ with $N = 5^{e-1}m$. Finally, $F = 20X + 17 = 17 + 5(4X)$ gives $S(5^e m) = 17U(5^e m) + 20T(5^e m) \in 5^e\mathbb{Z}_5$. □

6. Dwork crystals

We recall the results of Beukers and Vlasenko [2] that will be used; see also [3, 4]. Let p be an odd prime and let R be a p-adically complete integral domain of characteristic 0 with $\bigcap_s p^s R = 0$. Let $f \in R[x_1^{\pm 1}, \ldots, x_n^{\pm 1}]$ have Newton polytope $\Delta \subset \mathbb{R}^n$, and let $\mu = \Delta^\circ$ be its interior. We assume $\mu \cap \mathbb{Z}^n = \{u_0\}$ is a single point. Let $\Omega_f(\mu)$ be the R-module spanned by the rational functions $(m-1)!\, g/f^m$, where $m \geq 1$ and $g \in R[x^{\pm 1}]$ is supported in $m\mu$. Let $\widehat{\Omega}_f(\mu)$ be its p-adic completion. Derivations δ of R act coefficientwise and preserve $\widehat{\Omega}_f(\mu)$ [2, Prop. 3.4].

If ν is a vertex of Δ whose coefficient $f_{[\nu]}$ in f lies in $R^\times$, every element of $\widehat{\Omega}_f(\mu)$ has a formal Laurent expansion supported in the cone $C(\Delta - \nu)$ [2, §2]. A formal Laurent series $\sum a_k x^k$ is a *formal derivative*, i.e. lies in $\sum_i \theta_{x_i}(\cdot)$, if and only if $a_k \in \gcd(k_1, \ldots, k_n)R$ for all k [2, Lemma 2.2]. Let $U_f(\mu)$ be the submodule of elements whose expansion is a formal derivative. It does not depend on the choice of ν. Fix a Frobenius lift σ on R; all rings below carry one, e.g. $b \mapsto b^p$. Then by [2, Prop. 4.2] the submodule equals $\{\omega : C_p^s(\omega) \in p^s\widehat{\Omega}_{f^{\sigma^s}}(\mu)\ \forall s\}$, and the Cartier operator C_p is given by formulas independent of the vertex [2, §3, (4)–(7)]. The *Hasse–Witt invariant* $\beta_p \in R$ is the coefficient of $x^{(p-1)u_0}$ in f^{p-1}.

Theorem 6.1 (Beukers–Vlasenko [2, Theorems 4.3 and 5.7])**.** *Assume $\beta_p \in R^\times$ and put $\omega_0 = x^{u_0}/f$.*

- *(i) $\widehat{\Omega}_f(\mu)/U_f(\mu)$ is a free R-module of rank one with basis the class of ω_0. Hence for every derivation δ of R there is a unique $\eta_\delta \in R$ with $\delta(\omega_0) - \eta_\delta\,\omega_0 \in U_f(\mu)$.*
- *(ii) Let ν be a vertex of Δ with $f_{[\nu]} \in R^\times$, let $k \in C(\Delta - \nu) \cap \mathbb{Z}^n$, and let $a_m \in R$ be the coefficient of x^m in the expansion of ω_0 at ν. Then $\delta(a_{p^s k}) \equiv \eta_\delta\, a_{p^s k} \pmod{p^s R}$ for all $s \geq 1$.*

By the vertex-independence of $U_f(\mu)$, the element η_δ is the same for every vertex used in (ii). This is the key point of Section 9.

7. The unit-root crystal attached to S

Let p be an odd prime. In the variables (u, v, z, w) put

$$f'_b = (1+u+v)\big(1-uv(z+b/z)(w+1/w)\big).$$

Let $\mathrm{HW}'(b) \in \mathbb{Z}[b]$ be its Hasse–Witt polynomial (computed below), and let R_b be the p-adic completion of $\mathbb{Z}_p[b, b^{-1}, \mathrm{HW}'(b)^{-1}]$. This is a p-adically complete domain, since its reduction $\mathbb{F}_p[b, b^{-1}, \overline{\mathrm{HW}'}^{-1}]$ is a domain and it has no p-torsion.

Lemma 7.1. *(i) The Newton polytope Δ' of f'_b has the unique interior lattice point $u_0 = (1,1,0,0)$, and 0 is a vertex with coefficient 1.*
(ii) $\mathrm{HW}'(b) \equiv d_p(b) \pmod p$.
(iii) In the expansion of $\omega_0 = uv/f'_b$ at the vertex 0, the coefficient of $u^r v^s z^0 w^0$ is $[u^r v^s]\,\rho\, G_b(uv)$. In particular the coefficient of $(uv)^N = x^{Nu_0}$ is $d_N(b)$.

Proof. (i) Δ' is the Minkowski sum of the triangle $\mathrm{conv}\{0, e_u, e_v\}$ and the pyramid $\mathrm{conv}\{0, e_u+e_v \pm e_z \pm e_w\}$. So its points are $(\alpha+\mu, \beta+\mu, z, w)$ with $\alpha, \beta \ge 0$, $\alpha+\beta \le 1$, $0 \le \mu \le 1$ and $|z|, |w| \le \mu$. If (r,s,i,j) is an interior lattice point, moving in the z- or w-directions forces $|i|, |j| < 1$. Moving in the u-, v- and $(u+v)$-directions forces $r, s \ge 1$ and $r+s < 3$. Hence $(r,s,i,j) = (1,1,0,0)$, which is interior (take $\mu = 0.6$, $\alpha = \beta = 0.4$). All other monomials of f'_b have positive u- or v-degree, so 0 is a vertex.

(ii) The coefficient of $(uv)^{p-1} z^0 w^0$ in $f_b'^{\,p-1}$ is $\sum_l \binom{p-1}{2l} b^l \binom{2l}{l}^2 \cdot \frac{(p-1)!}{a!^2 (p-1-2a)!}$ with $a = p-1-2l$. Modulo p, $\binom{p-1}{2l} \equiv 1$, and the multinomial is $\binom{p-1}{2a}\binom{2a}{a} \equiv \binom{2a}{a}$ for $2a \le p-1$. When $2a > p-1$ the multinomial vanishes and so does $\binom{2a}{a} \bmod p$. This gives $d_p(b)$.

(iii) $1/f'_b = (1+u+v)^{-1} \sum_j (uv)^j (z+b/z)^j (w+1/w)^j$, and $\mathrm{CT}_{z,w}(z+b/z)^j(w+1/w)^j = b^{j/2}\binom{j}{j/2}^2$ for even j (and 0 otherwise). Also $[u^i v^i](1+u+v)^{-1} = \binom{2i}{i}$. ☐

By Lemma 7.1, Theorem 6.1 applies to f'_b over R_b. Let $\eta_b \in R_b$ denote η_δ for $\delta = \theta_b$:

$$\theta_b\Big(\frac{uv}{f'_b}\Big) \equiv \eta_b \frac{uv}{f'_b} \pmod{U_{f'_b}(\mu)}. \tag{5}$$

Proposition 7.2. *Let $p \equiv 1, 3, 7, 9 \pmod{20}$. Then evaluation $b \mapsto -4$ extends to a continuous ring homomorphism $R_b \to \mathbb{Z}_p$. If $20\,\eta_b(-4) + 7 = 0$, then $p^{v_p(N)} \mid S(N)$ for every $N \ge 1$.*

Proof. $\mathrm{HW}'(-4) \equiv d_p(-4)$ is a unit by Corollary 4.3, which gives the evaluation map. Let $\Omega = (20\theta_b + 7)(uv/f'_b) \in \widehat{\Omega}_{f'_b}(\mu)$. By (5), $\Omega - (20\eta_b + 7)uv/f'_b \in U_{f'_b}(\mu)$. So every coefficient $a_k(b) \in R_b$ of its expansion at 0 lies in $\gcd(k)R_b$, and hence $a_k(-4) \in \gcd(k)\mathbb{Z}_p$. Since $20\eta_b(-4) + 7 = 0$, the coefficients of $\Omega|_{b=-4}$ lie in $\gcd(k)\mathbb{Z}_p$. By Lemma 7.1(iii), the coefficient of $u^r v^s z^0 w^0$ in $\Omega|_{b=-4}$ is $[u^r v^s]\,\rho\,((20\theta_b+7)G_b)(uv)|_{b=-4} = [u^r v^s]\rho\Phi(uv)$, and $\gcd(r,s,0,0) = \gcd(r,s)$. Conclude with Corollary 2.5. ☐

8. Ramanujan's crystal and Sun's theorem

Let $\mathcal{F}(T) = \sum_{k \ge 0} A_k T^k$, $\mathcal{F}_{<m}(T) = \sum_{k<m} A_k T^k$, and $\eta_{\mathcal{R}} = \theta_T \log \mathcal{F}(T) \in \mathbb{Z}_p[[T]]$. Let R_T be the p-adic completion of $\mathbb{Z}_p[T, \mathcal{F}_{<p}(T)^{-1}]$, with Frobenius lift $\sigma(T) = T^p$. Since $\mathcal{F}_{<p}(0) = 1$, expansion at $T = 0$ gives a map $R_T \to \mathbb{Z}_p[[T]]$. It is injective and p-saturated: if the expansion of r lies in $p^s\mathbb{Z}_p[[T]]$ then $r \in p^s R_T$. Indeed, approximate r modulo p^s by $g/\mathcal{F}_{<p}^k$ with $g \in \mathbb{Z}_p[T]$; then $g \in p^s\mathbb{Z}_p[T]$ because $\mathcal{F}_{<p}$ is a unit power series.

Proposition 8.1. *(i) $\eta_{\mathcal{R}} \in R_T$, and $\eta_{\mathcal{R}} \equiv \theta_T \log \mathcal{F}_{<p^s} \pmod{p^s R_T}$ for all $s \ge 1$. (ii) If $p \equiv 1, 3, 7, 9 \pmod{20}$, then evaluation at $t_0 = -1/1024$ extends to $R_T \to \mathbb{Z}_p$, and $\eta_{\mathcal{R}}(t_0) = -\frac{3}{20}$.*

Proof. (i) We have $A_k = \mathrm{CT}(g^k)$ for $g = (1+x_1+x_2+x_3)^4/(x_1x_2x_3)$, whose Newton polytope has 0 as its only interior lattice point. By Dwork's congruences in the form of Mellit–Vlasenko [11] (see also [3, §1]), $\frac{\mathcal{F}(T)}{\mathcal{F}(T^p)} \equiv \frac{\mathcal{F}_{<p^s}(T)}{\mathcal{F}_{<p^{s-1}}(T^p)} \pmod{p^s\mathbb{Z}_p[[T]]}$ for all $s \geq 1$. Lucas congruences give $\mathcal{F}_{<p^s} \equiv \mathcal{F}_{<p}^{(p^s-1)/(p-1)} \pmod p$, so the right sides are units of R_T. By saturation they form a Cauchy sequence in R_T, whose limit λ has expansion $\mathcal{F}(T)/\mathcal{F}(T^p)$. The formal identity $\eta_{\mathcal{R}} = \theta_T \log\lambda + p\,\sigma(\eta_{\mathcal{R}})$ gives $\eta_{\mathcal{R}} = \sum_{i\geq 0} p^i\sigma^i(\theta_T\log\lambda) \in R_T$.

For the congruence, put $D_s = \theta_T \log \mathcal{F}_{<p^s} - \eta_{\mathcal{R}} \in R_T$, with $D_0 = -\eta_{\mathcal{R}}$. Taking $\theta_T\log$ of Dwork's congruence (a congruence of units modulo p^s) gives $D_s \equiv p\,\sigma(D_{s-1}) \pmod{p^s}$. Hence by induction $D_s \in p^s R_T$.

(ii) The evaluation exists because $\mathcal{F}_{<p}(t_0) \in \mathbb{Z}_p^\times$ (Corollary 4.3). Sun proved [15, Theorem 1.2, (1.5)] that for every $n \geq 1$,

$$4(2n+1)\binom{2n}{n} \;\Big|\; \sum_{k=0}^{n}(20k+3)\binom{2k}{k}^2\binom{4k}{2k}(-2^{10})^{n-k}.$$

Take $n = p^s-1$. Adding $(p^s-1)+(p^s-1)$ in base p produces s carries, so $v_p\binom{2p^s-2}{p^s-1} = s$ by Kummer's theorem. Dividing by the unit $(-1024)^{p^s-1}$, $\sum_{k<p^s}(20k+3)A_k t_0^k = \big((20\theta_T+3)\mathcal{F}_{<p^s}\big)(t_0) \equiv 0 \pmod{p^s}$. Since $\mathcal{F}_{<p^s}(t_0)$ is a unit, part (i) gives $20\eta_{\mathcal{R}}(t_0)+3 \equiv 0 \pmod{p^s}$ for every s. □

9. Identification of the connection

Theorem 9.1. *Let p be an odd prime. The substitution $T \mapsto 1/(256b)$ defines a ring homomorphism $R_T \to R_b$ compatible with $\theta_b = -\theta_T$, and in R_b*

$$\eta_b(b) = -\frac{1}{2} - \eta_{\mathcal{R}}\Big(\frac{1}{256b}\Big).$$

The substitution is well defined because $\mathcal{F}_{<p}(1/(256b)) \equiv b^{-h}d_p(b) \equiv b^{-h}\mathrm{HW}'(b) \pmod p$ (Proposition 4.1 and Lemma 7.1), which is a unit of R_b. The proof of the identity occupies the rest of this section.

9.1. **A diagonal model.** In the variables (x,z,w,y) put $g(z,w) = (z+1/z)(w+1/w)$, $L(y) = y+2+1/y$ and

$$f_c = \big(1 - c\,x\,g(z,w)\big)\big(1 - x\,L(y)\big).$$

Let $\mathrm{HW}_c(c)$ be its Hasse–Witt polynomial and R_c the p-adic completion of $\mathbb{Z}_p[c,c^{-1},\mathrm{HW}_c(c)^{-1}]$.

Lemma 9.2. *(i) The Newton polytope Δ of f_c has the unique interior lattice point $e_x = (1,0,0,0)$. Both 0 (coefficient 1) and $\nu_1 = (2,-1,-1,-1)$ (coefficient c) are vertices.*
(ii) $\mathrm{HW}_c(c) \equiv d_p(c^2) \pmod p$. Its leading coefficient in c is $\binom{p-1}{h}^2 \equiv 1$.
(iii) The map $\iota : R_b \to R_c$, $b \mapsto c^2$, is an injective ring homomorphism with $\theta_c \circ \iota = 2\,\iota \circ \theta_b$.
(iv) In the expansion of $\omega_1 = x/f_c$ at 0, the coefficient of $x^N z^0 w^0 y^0$ is $d_N(c^2)$.

Proof. (i) Δ is the Minkowski sum of the pyramid $\{(\mu,z,w) : 0 \leq \mu \leq 1,\ |z|,|w| \leq \mu\}$ and the triangle $\{(\nu,y) : 0 \leq \nu \leq 1,\ |y| \leq \nu\}$. An interior lattice point (r,i,j,l) must have $|i|,|j|,|l| < 1$ and $0 < r < 2$, so it is e_x, which is interior (take $\mu = \nu = \frac12$). The functional $e_x - e_z - e_w - e_y$ attains its maximum on the two summands uniquely, at $e_x - e_z - e_w$ and $e_x - e_y$. So ν_1 is a vertex, and its coefficient is $(-c)(-1) = c$.

(ii) The coefficient of x^{p-1} in f_c^{p-1} is $\sum_{a \text{ even}} \binom{p-1}{a} c^a \binom{a}{a/2}^2 \binom{p-1}{e}(-1)^e\binom{2e}{e}$ with $e = p-1-a$. Reduce modulo p as in Lemma 7.1(ii). The top term $a = p-1$ gives the leading coefficient.

(iii) $\mathrm{HW}'(c^2) \equiv \mathrm{HW}_c(c) \pmod p$, so it is a unit in R_c. Injectivity holds modulo p (a map of localized polynomial rings over $\mathbb{F}_p$), hence on completions.

(iv) The expansion at 0 is $x\sum_j (cxg + xL - cx^2gL)^j = x\,(1-cxg)^{-1}(1-xL)^{-1}$. Here $\mathrm{CT}_{z,w}(1-cxg)^{-1} = \sum_j c^j x^j\,\mathrm{CT}(g^j) = G_{c^2}(x)$ and $\mathrm{CT}_y(1-xL)^{-1} = E(x)$. □

Let $\eta_c \in R_c$ be the connection coefficient of f_c for $\delta = \theta_c$ (Theorem 6.1(i)).

Lemma 9.3. $\eta_c = 2\,\iota(\eta_b)$.

Proof. Apply Theorem 6.1(ii) at the vertex 0 to f'_b with $k = u_0$, and to f_c with $k = e_x$. By Lemmas 7.1(iii) and 9.2(iv), $\theta_b d_{p^s}(b) \equiv \eta_b d_{p^s}(b)$ in R_b and $\theta_c\, d_{p^s}(c^2) \equiv \eta_c\, d_{p^s}(c^2)$ in R_c, both modulo p^s. Apply ι to the first and use $\theta_c \circ \iota = 2\iota \circ \theta_b$. Since $d_{p^s}(c^2)$ is a unit (Lemma 4.4), $\eta_c \equiv 2\iota(\eta_b) \pmod{p^s}$ for all s. □

9.2. **The corner expansion.**

Lemma 9.4. *Let $k_1 = e_x - \nu_1 = (-1,1,1,1) \in C(\Delta - \nu_1)$ and $m = p^s$, $h_s = (m-1)/2$. In the expansion of ω_1 at ν_1, the coefficient of $x^{mk_1} = x^{-m}z^m w^m y^m$ is*

$$\alpha_s(c) = \sum_{l=0}^{h_s} c^{-1-2l} \binom{h_s + l}{2l}^2 \binom{2m-1-2l}{2l}.$$

Proof. Write $(1-cxg)(1-xL) = cxg\cdot xL\cdot(1-(cxg)^{-1})(1-(xL)^{-1})$, with $g = (1+z^2)(1+w^2)/(zw)$ and $L = (1+y)^2/y$. This gives

$$f_c = c\,x^{\nu_1}\,(1+z^2)(1+w^2)(1+y)^2\Big(1 - \frac{zw}{cx(1+z^2)(1+w^2)}\Big)\Big(1 - \frac{y}{x(1+y)^2}\Big).$$

The monomials $z^2, w^2, y, y^2, zw/x, y/x$ occurring in the last five factors lie in $(\Delta - \nu_1)\setminus\{0\}$. So these factors are units of the ring of Laurent series supported in the pointed cone $C(\Delta - \nu_1)$, and the expansion of ω_1 at ν_1 is

$$\omega_1 = \frac{zwy}{cx} \sum_{\alpha,\beta \ge 0} \frac{(zw)^\alpha}{(cx)^\alpha} \frac{y^\beta}{x^\beta} (1+z^2)^{-1-\alpha}(1+w^2)^{-1-\alpha}(1+y)^{-2-2\beta}.$$

The coefficient of $x^{-m}z^m w^m y^m$ requires $\alpha + \beta = m-1$. The z- and w-parts each give $\binom{-1-\alpha}{(m-1-\alpha)/2}$, and the y-part gives $\binom{-2-2\beta}{\alpha}$. With $\alpha = 2l$ (forced, as m is odd), these equal $\pm\binom{h_s+l}{2l}$ and $\binom{2m-1-2l}{2l}$, and the signs square away. □

Lemma 9.5. *For fixed $l \ge 0$, as $s \to \infty$, $\binom{h_s+l}{2l}^2\binom{2p^s-1-2l}{2l} \to \binom{4l}{2l}\binom{2l}{l}^2 256^{-l} = A_l 256^{-l}$ in $\mathbb{Z}_p$.*

Proof. Both binomials are polynomials in p^s with coefficients in $\frac{1}{(2l)!}\mathbb{Z}[\frac12]$, and $h_s \to -\frac12$, $2p^s - 1 - 2l \to -1 - 2l$. The limits are $\binom{l-1/2}{2l}^2 = \binom{2l}{l}^2/256^l$ (see the proof of Proposition 4.1) and $\binom{-1-2l}{2l} = \binom{4l}{2l}$. □

9.3. **Proof of Theorem 9.1.** Let $u = 1/c$. Let $\mathfrak{L}$ be the ring of two-sided series $\sum_{n\in\mathbb{Z}} e_n u^n$ with $e_n \in \mathbb{Z}_p$ and $e_n \to 0$ as $n \to -\infty$. (Products converge, since in each coefficient of a product one factor's index is $\le n/2$.) Expansion at $u = 0$ defines a ring homomorphism $\exp : \mathbb{Z}_p[c, c^{-1}, \mathrm{HW}_c^{-1}] \to \mathfrak{L}$, since HW_c has unit leading coefficient. It maps p^s-multiples to p^s-multiples, so it extends continuously to R_c, it commutes with $\theta_c = -\theta_u$, and it is injective by the saturation argument of Section 8. On the image of $R_T \to R_c$, $T \mapsto 1/(256c^2)$, it is given by $T \mapsto u^2/256$ applied to expansions at $T = 0$.

By Theorem 6.1(ii) at the vertex ν_1 (coefficient $c \in R_c^\times$) and Lemma 9.4,

$$\theta_c\,\alpha_s - \eta_c\,\alpha_s \in p^s R_c, \qquad \text{hence} \qquad \theta_c\,\alpha_s - \exp(\eta_c)\,\alpha_s \in p^s\mathfrak{L} \quad (s \ge 1).$$

Here $\alpha_s \in u\mathbb{Z}[u]$. Let $\Pi(u) = u\,\mathcal{F}(u^2/256) = \sum_l A_l 256^{-l} u^{2l+1}$. By Lemma 9.5, $\alpha_s \to \Pi$ coefficientwise. Write $\exp(\eta_c) = \sum e_n u^n$. The coefficient of u^n in $\exp(\eta_c)\alpha_s$ is $\sum_{j\ge1} e_{n-j}[u^j]\alpha_s$, with

$|[u^j]\alpha_s| \le 1$ and $e_{n-j} \to 0$ as $j \to \infty$. So it converges to $\sum_j e_{n-j}[u^j]\Pi$ as $s \to \infty$. Letting $s \to \infty$ gives $\theta_c\Pi = \exp(\eta_c)\Pi$ in $\mathfrak{L}$. Since $\Pi \in u(1+u^2\mathbb{Z}_p[[u]])$ is invertible in $u^{-1}\mathbb{Z}_p[[u]] \subset \mathfrak{L}$,

$$\exp(\eta_c) = \frac{\theta_c\Pi}{\Pi} = -\theta_u \log \Pi = -1 - 2\,(\theta_T \log \mathcal{F})(u^2/256) = \exp\Big(-1 - 2\,\eta_{\mathcal{R}}\big(\tfrac{1}{256c^2}\big)\Big).$$

By injectivity, $\eta_c = -1 - 2\eta_{\mathcal{R}}(1/(256c^2))$ in R_c. Since ι maps $\eta_{\mathcal{R}}(1/(256b))$ to $\eta_{\mathcal{R}}(1/(256c^2))$, Lemma 9.3 turns this into

$$\iota\Big(\eta_b + \frac{1}{2} + \eta_{\mathcal{R}}\big(\tfrac{1}{256b}\big)\Big) = 0.$$

Injectivity of ι finishes the proof. □

10. Proof of the Main Theorem

Proof of Theorem 1.1. Let $p \equiv 1,3,7,9 \pmod{20}$. Evaluating Theorem 9.1 at $b = -4$, which corresponds to $T = t_0 = -1/1024$, and using Proposition 8.1(ii), we get $\eta_b(-4) = -\frac{1}{2} + \frac{3}{20} = -\frac{7}{20}$. So $20\eta_b(-4) + 7 = 0$, and Proposition 7.2 gives $p^{v_p(N)} \mid S(N)$ for all N. □

Proof of the Main Theorem. Let $n \ge 1$ and $p \mid n$, and write $n = p^e m$ with $p \nmid m$. For $p = 2$ use Theorem 3.1; for $p = 5$, Theorem 5.6; for $p \equiv 11,13,17,19 \pmod{20}$, Theorem 5.5; for $p \equiv 1,3,7,9 \pmod{20}$, Theorem 1.1. Every prime is in exactly one of these classes, so $p^{v_p(n)} \mid S(n)$ for all p, i.e. $n \mid S(n)$. □

Remark 10.1. The proof gives more. (a) For every prime p with $p \equiv 1,3,7,9 \pmod{20}$, $\rho\Phi(uv)$ is a p-adic formal divergence; equivalently, $p^{\min(v_p r, v_p s)} \mid [u^r v^s]\rho\Phi(uv)$. In particular the diagonal coefficients $Y_2(n) = [u^n v^n]\rho\Phi(uv) = \sum_l (20l+7)\gamma_l \binom{2i}{i}$, $i = n-1-2l$, satisfy $p^{v_p(n)} \mid Y_2(n)$; compare the diagonal criterion of [10, §8]. (b) For every odd prime and every $b \in \mathbb{Z}_p^\times$ with $d_p(b) \not\equiv 0$, Theorem 9.1 evaluated at b expresses the unit-root connection of f_b' through Ramanujan's family at $1/(256b)$.

11. Discussion

Where the new input lies. Proposition 2.2 and Sections 7, 8 and 9 are new. Proposition 2.3 is a variant of the negative-index reduction of [10]. Sections 3 and 5 reproduce arguments of [10], and the Hasse-invariant computation of Section 4 follows [10]. We know of no earlier proof of the conjecture. Searches of arXiv and of the literature on Sun's divisibility conjectures (including [15, 13, 9]) found the problem listed as open and no proof. Such a search cannot by itself establish priority. The essential new step is Theorem 9.1. It compares two unit-root crystals that are *not* related by any monomial change of variables: the crystal of S, and Ramanujan's crystal at the reciprocal parameter. Their Hasse invariants agree modulo p (Proposition 4.1), but that alone does not determine a unit root. The comparison comes from the corner expansion of the diagonal model, whose Hasse–Witt coefficients converge to Ramanujan's period. Conceptually this reflects the inversion $b \leftrightarrow 1/b$ of the hypergeometric equation in Remark 1.2, and the Landen-type involution $z \mapsto (1-iz)/(z-i)$, which satisfies $(z+z^{-1})(\mu(z)+\mu(z)^{-1}) = 4$ and exchanges the two degenerations of the Legendre family. No monomial substitution can realise this involution, which is why the comparison has to go through a limit.

The role of the CM point. The prime p is split (ordinary) or inert (supersingular) according to the reduction of the CM curve with $j = j(\sqrt{-5})$. At inert primes the three moments vanish and the digit operator of Section 5 suffices. At split primes the moments are units, and the divisibility is equivalent to the precise p-adic value $\eta_b(-4) = -7/20$. That value is the p-adic counterpart of Ramanujan's formula, supplied by Sun's WZ theorem through Proposition 8.1.

Verification. The Appendix contains an exact-arithmetic script. It checks the conjecture for $n \le 2000$ and every identity, congruence and hypothesis used above on explicit ranges: the decomposition of Proposition 2.2; the polytope facts; the Hasse–Witt congruences; the agreement of the two vertex expansions predicted by Theorem 6.1(ii); Theorem 9.1 at the level of truncations (82 cases); Sun's theorem; the CM dichotomy for $p < 1500$; the moments at inert primes and at $p = 5$; the Legendre–Clausen identity; and the two-variable divisibility of $\rho\Phi(uv)$. Earlier exact runs verified $n \mid S(n)$ for all $n \le 10^6$ [7]. No step of the proof depends on these computations.

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Appendix A. Verification script and output

The following Python 3 script (standard library only) was run as `python3 verify_full.py` (about one minute). Its output is reproduced after the listing.

```
#!/usr/bin/env python3
"""Exact-arithmetic checks for the proof of Sun's A179524 conjecture.

Run:  python3 verify_full.py            (about one to three minutes)
Every check prints PASS or FAIL; the script exits with status 1 on any FAIL.
Nothing here is used as a proof step; the checks confirm each identity,
congruence and hypothesis used in the paper on explicit ranges.
"""
from math import comb, factorial, gcd
from fractions import Fraction as Fr
import sys

FAILED = []
```

```
def report(name, ok):
    print(("PASS  " if ok else "FAIL  ") + name)
    if not ok:
        FAILED.append(name)


def primes(n):
    s = bytearray([1]) * (n + 1)
    s[0:2] = b"\x00\x00"
    for i in range(2, int(n ** 0.5) + 1):
        if s[i]:
            s[i * i::i] = bytearray(len(s[i * i::i]))
    return [i for i in range(n + 1) if s[i]]


def gam(l):                       # gamma_l = (-4)^l C(2l,l)^2
    return (-4) ** l * comb(2 * l, l) ** 2


def A(k):                         # Ramanujan coefficients (4k)!/k!^4
    return factorial(4 * k) // factorial(k) ** 4


def a(n):                         # a_n = sum_k (-4)^k C(n,2k)^2 C(2k,k)^2
    return sum(gam(k) * comb(n, 2 * k) ** 2 for k in range(n // 2 + 1))


def vp(x, p):
    if x == 0:
        return 10 ** 9
    v = 0
    while x % p == 0:
        x //= p
        v += 1
    return v


# 1. Sun's conjecture n | S(n), exact, n <= NMAX
NMAX = 2000
S, ok = 0, True
for n in range(1, NMAX + 1):
    S += (20 * (n - 1) + 17) * a(n - 1)
    if S % n:
        ok = False
report("n | S(n) for 1 <= n <= %d" % NMAX, ok)

# 2. Telescoping identity and the decomposition S = 10 H3 + H2 (Prop. 2.4)
ok = True
for m in range(0, 14):
    for N in range(0, 60):
        lhs = sum((2 * n - m + 1) * comb(n, m) ** 2 for n in range(N))
        rhs = N * comb(N, m + 1) * comb(N - 1, m) if N >= 1 else 0
        ok &= (lhs == rhs)
for N in range(1, 300):
    S = sum((20 * n + 17) * a(n) for n in range(N))
    H3 = N * sum(gam(k) * comb(N, 2 * k + 1) * comb(N - 1, 2 * k) for k in range(N))
    H2 = sum((20 * k + 7) * gam(k) * comb(n, 2 * k) ** 2
             for n in range(N) for k in range(n // 2 + 1))
    ok &= (S == 10 * H3 + H2)
report("telescoping identity and S = 10*H3 + H2 (N < 300)", ok)

# 3. Unique interior lattice points of the two Newton polytopes (Lemma 6.4),
#    via the explicit description used in the proof.
```

```
def interior_fprime(r, s, i, j):   # exists mu: max(|i|,|j|,(r+s-1)/2) < mu < min(r,s,1)
    lo = max(abs(i), abs(j), Fr(r + s - 1, 2))
    hi = min(r, s, 1)
    return lo < hi


def interior_fc(r, i, j, l):       # exists mu,nu in (0,1): mu+nu=r, mu>max|i|,|j|, nu>|l|
    lo_mu = max(abs(i), abs(j), r - 1)
    hi_mu = min(1, r - abs(l))
    return lo_mu < hi_mu
pts1 = [(r, s, i, j) for r in range(0, 3) for s in range(0, 3)
        for i in range(-2, 3) for j in range(-2, 3) if interior_fprime(r, s, i, j)]
pts2 = [(r, i, j, l) for r in range(0, 3) for i in range(-2, 3)
        for j in range(-2, 3) for l in range(-2, 3) if interior_fc(r, i, j, l)]
report("interior lattice points: f'_b -> %s, f_c -> %s" % (pts1, pts2),
       pts1 == [(1, 1, 0, 0)] and pts2 == [(1, 0, 0, 0)])


def d_coeffs(n):                # d_n(b) = sum_l b^l C(2l,l)^2 C(2i,i), i = n-1-2l
    return {l: comb(2 * l, l) ** 2 * comb(2 * (n - 1 - 2 * l), n - 1 - 2 * l)
            for l in range((n - 1) // 2 + 1)}


# 4. Hasse-Witt congruences (Lemma 6.4)
ok = True
for p in primes(60)[1:]:
    hw1 = {}
    for l in range((p - 1) // 2 + 1):
        aa = p - 1 - 2 * l
        if 2 * aa <= p - 1:
            mult = factorial(p - 1) // (factorial(aa) ** 2 * factorial(p - 1 - 2 * aa))
            hw1[l] = comb(p - 1, 2 * l) * comb(2 * l, l) ** 2 * mult
    d = d_coeffs(p)
    ok &= all((hw1.get(l, 0) - d.get(l, 0)) % p == 0 for l in set(hw1) | set(d))
    hw2 = {}
    for a2 in range(0, p, 2):
        cc = p - 1 - a2
        hw2[a2] = (comb(p - 1, a2) * comb(a2, a2 // 2) ** 2 * comb(p - 1, cc)
                   * (-1) ** cc * comb(2 * cc, cc))
    ok &= all((hw2.get(e, 0) - (d.get(e // 2, 0) if e % 2 == 0 else 0)) % p == 0
              for e in set(hw2) | {2 * l for l in d})
report("HW(f'_b) == d_p(b) and HW(f_c) == d_p(c^2) mod p, 3 <= p < 60", ok)


def ev(P, x, mod, der=False):
    return sum((e if der else 1) * cf * pow(x, e, mod) for e, cf in P.items()) % mod


def corner(m):                  # corner-vertex coefficients a_{p^s k_1}(c), exponents of c
    h = (m - 1) // 2
    return {-1 - 2 * l: comb(h + l, 2 * l) ** 2 * comb(2 * m - 1 - 2 * l, 2 * l)
            for l in range(h + 1)}


# 5. Two vertex expansions give the same connection (BV Thm 5.7 consistency)
ok, tested = True, 0
for p, s in [(3, 3), (3, 4), (7, 2), (7, 3), (11, 2), (13, 2), (23, 2)]:
    M = p ** s
    mod = M
    cr = corner(M)
    dd = {2 * l: v for l, v in d_coeffs(M).items()}
    for cv in range(2, 14):
        if cv % p == 0:
            continue
        X, D = ev(cr, cv, mod), ev(dd, cv, mod)
```

```
        if X % p == 0 or D % p == 0:
            continue
        e1 = ev(cr, cv, mod, True) * pow(X, -1, mod) % mod
        e2 = ev(dd, cv, mod, True) * pow(D, -1, mod) % mod
        ok &= (e1 == e2)
        tested += 1
report("theta log(corner) == theta log(diagonal) mod p^s (%d cases)" % tested, ok)

# 6. Key identity eta_b(b) = -1/2 - eta_R(1/(256 b)) at the level of truncations
ok, tested = True, 0
for p, s in [(3, 4), (7, 3), (11, 2), (13, 2), (23, 2), (29, 2)]:
    M = p ** s
    mod = M
    dd = d_coeffs(M)
    Fc = {k: A(k) for k in range(M)}
    for bv in range(-6, 12):
        if bv % p == 0:
            continue
        D = ev(dd, bv % mod, mod)
        t = pow(256 * bv % mod, -1, mod)
        F = ev(Fc, t, mod)
        if D % p == 0 or F % p == 0:
            continue
        eta = ev(dd, bv % mod, mod, True) * pow(D, -1, mod) % mod
        etaR = ev(Fc, t, mod, True) * pow(F, -1, mod) % mod
        ok &= ((eta + etaR) * 2 + 1) % mod == 0
        tested += 1
report("theta log d_{p^s}(b) + theta log F_{<p^s}(1/(256b)) == -1/2 mod p^s (%d cases)"
       % tested, ok)

# 7. Sun's theorem (EJC 2013, (1.5)) and the instance n = p^s - 1
ok = all(sum((20 * k + 3) * A(k) * (-1024) ** (n - k) for k in range(n + 1))
         % (4 * (2 * n + 1) * comb(2 * n, n)) == 0 for n in range(1, 300))
report("4(2n+1)C(2n,n) | sum_{k<=n}(20k+3)A_k(-1024)^{n-k}, 1 <= n < 300", ok)
ok = True
for q in [3, 9, 27, 81, 7, 49, 23, 29, 41, 43]:
    tot = sum((20 * k + 3) * A(k) * (-1024) ** (q - 1 - k) for k in range(q))
    p = [r for r in primes(q) if q % r == 0][0]
    ok &= (tot % q == 0) and vp(comb(2 * q - 2, q - 1), p) == vp(q, p)
report("p^s | sum_{k<p^s}(20k+3)A_k(-1024)^{p^s-1-k} and v_p C(2p^s-2,p^s-1) = s", ok)

# 8. Ordinary / supersingular dichotomy of the Hasse invariant d_p(-4)
ok = True
for p in primes(1500)[1:]:
    hv = ev(d_coeffs(p), -4 % p, p)
    if p % 20 in (1, 3, 7, 9):
        ok &= hv != 0
    else:
        ok &= hv == 0
report("d_p(-4) != 0 mod p iff p = 1,3,7,9 mod 20 (odd p < 1500)", ok)

# 9. Moments at inert primes and the p = 5 table
def f4(r, s):
    return (-4) ** s * (factorial(r) // (factorial(s) ** 2 * factorial(r - 2 * s))) ** 2
ok = True
for p in primes(400):
    if p % 20 in (11, 13, 17, 19):
        U = sum(f4(r, s) for r in range(p) for s in range(r // 2 + 1))
        T = sum(r * f4(r, s) for r in range(p) for s in range(r // 2 + 1))
        V = sum(s * f4(r, s) for r in range(p) for s in range(r // 2 + 1))
        ok &= (U % p == 0 and T % p == 0 and V % p == 0)
report("U(p), T(p), V(p) == 0 mod p for inert p < 400", ok)
p = 5
H = [Fr(0)] + [sum(Fr(1, i) for i in range(1, r + 1)) for r in range(1, p)]
Q = Fr(4 ** 4 - 1, 5)
```

```
al = sum(2 * f4(r, s) * (H[r] - H[r - 2 * s]) for r in range(p) for s in range(r // 2 + 1))
be = sum(f4(r, s) * (Q + 4 * H[r - 2 * s] - 4 * H[s]) for r in range(p) for s in range(r // 2 + 1))
U5 = sum(f4(r, s) for r in range(p) for s in range(r // 2 + 1))


def zero_mod5(x):
    return (x.numerator * pow(x.denominator, -1, 5)) % 5 == 0
report("p = 5: U(5) = %d, alpha = %s, beta = %s, both == 0 mod 5" % (U5, al, be),
       U5 == -155 and zero_mod5(al) and zero_mod5(be))

# 10. Legendre-Clausen identity and the Hasse invariant (Prop. 4.2)
def poly_mul(P, Q2):
    R2 = {}
    for i, x in P.items():
        for j, y in Q2.items():
            R2[i + j] = R2.get(i + j, 0) + x * y
    return R2


def legendre(nn):                # P_n(t) via Bonnet recursion, rational coefficients in t
    P0, P1 = {0: Fr(1)}, {1: Fr(1)}
    if nn == 0:
        return P0
    for k in range(1, nn):
        P2 = {}
        for e, cf in P1.items():
            P2[e + 1] = P2.get(e + 1, 0) + Fr(2 * k + 1, k + 1) * cf
        for e, cf in P0.items():
            P2[e] = P2.get(e, 0) - Fr(k, k + 1) * cf
        P0, P1 = P1, P2
    return P1
ok = True
for dd_ in range(0, 15):
    L = legendre(dd_)
    lhs = poly_mul(L, L)
    rhs = {}
    for j in range(dd_ + 1):
        base = {2: Fr(1, 4), 0: Fr(-1, 4)}
        term = {0: Fr(comb(dd_, j) * comb(dd_ + j, j) * comb(2 * j, j))}
        for _ in range(j):
            term = poly_mul(term, base)
        for e, cf in term.items():
            rhs[e] = rhs.get(e, 0) + cf
    ok &= all(lhs.get(e, 0) == rhs.get(e, 0) for e in set(lhs) | set(rhs))
report("P_d(t)^2 = sum_j C(d,j)C(d+j,j)C(2j,j)((t^2-1)/4)^j, d < 15", ok)
ok = True
for p in primes(60)[3:]:
    h, d = (p - 1) // 2, p // 4
    # coefficient of x^{p-1} in (x(x^2+2x+b'))^h, b'=(1-t)/2, as polynomial in t
    co = {}
    for k in range(d + 1):
        # choose k factors b'x, (h-2k) factors 2x^2, k factors x^3: h!/(k!(h-2k)!k!)
        mult = factorial(h) // (factorial(k) * factorial(h - 2 * k) * factorial(k))
        term = {0: Fr(mult * 2 ** (h - 2 * k))}
        for _ in range(k):
            term = poly_mul(term, {0: Fr(1, 2), 1: Fr(-1, 2)})
        for e, cf in term.items():
            co[e] = co.get(e, 0) + cf
    L = legendre(d)
    for e in set(co) | set(L):
        x = co.get(e, 0) - 2 ** h * L.get(e, 0)
        ok &= (x.numerator * pow(x.denominator, -1, p)) % p == 0
report("Hasse invariant of E_t == 2^h P_d(t) mod p, 7 <= p < 60", ok)

# 11. The two-variable gcd criterion for rho*F(uv) (consequence of the theorem)
```

```
ok = True
for r in range(1, 121):
    for s in range(1, 121):
        val = 0
        for l in range((min(r, s) - 1) // 2 + 1):
            aa, bb = r - 1 - 2 * l, s - 1 - 2 * l
            val += (20 * l + 7) * gam(l) * (-1) ** (aa + bb) * comb(aa + bb, aa)
        ok &= (val % gcd(r, s) == 0)
report("gcd(r,s) | [u^r v^s] rho F(uv) for r, s <= 120", ok)

print()
print("ALL CHECKS PASSED" if not FAILED else "FAILED: " + "; ".join(FAILED))
sys.exit(1 if FAILED else 0)
```

Output:

```
PASS  n | S(n) for 1 <= n <= 2000
PASS  telescoping identity and S = 10*H3 + H2 (N < 300)
PASS  interior lattice points: f'_b -> [(1, 1, 0, 0)], f_c -> [(1, 0, 0, 0)]
PASS  HW(f'_b) == d_p(b) and HW(f_c) == d_p(c^2) mod p, 3 <= p < 60
PASS  theta log(corner) == theta log(diagonal) mod p^s (62 cases)
PASS  theta log d_{p^s}(b) + theta log F_{<p^s}(1/(256b)) == -1/2 mod p^s (82 cases)
PASS  4(2n+1)C(2n,n) | sum_{k<=n}(20k+3)A_k(-1024)^{n-k}, 1 <= n < 300
PASS  p^s | sum_{k<p^s}(20k+3)A_k(-1024)^{p^s-1-k} and v_p C(2p^s-2,p^s-1) = s
PASS  d_p(-4) != 0 mod p iff p = 1,3,7,9 mod 20 (odd p < 1500)
PASS  U(p), T(p), V(p) == 0 mod p for inert p < 400
PASS  p = 5: U(5) = -155, alpha = 1440, beta = -37270/3, both == 0 mod 5
PASS  P_d(t)^2 = sum_j C(d,j)C(d+j,j)C(2j,j)((t^2-1)/4)^j, d < 15
PASS  Hasse invariant of E_t == 2^h P_d(t) mod p, 7 <= p < 60
PASS  gcd(r,s) | [u^r v^s] rho F(uv) for r, s <= 120

ALL CHECKS PASSED
```